Content-Centric Modelling of Rage-Bait Lifecycles on Social Media with Toxicity Feedback

APM348 — 10-Minute Presentation

Neel Sarkar Piper Scott

University of Toronto

April 1, 2026

Why This Matters

Engagement-optimized algorithms amplify provocative content. This boosts short-term engagement but increases toxicity and drives users away [1, 2].

Central question

How does platform amplification shape the full lifecycle of rage-bait, and what are the costs for toxicity and user retention?

Three things we will show:

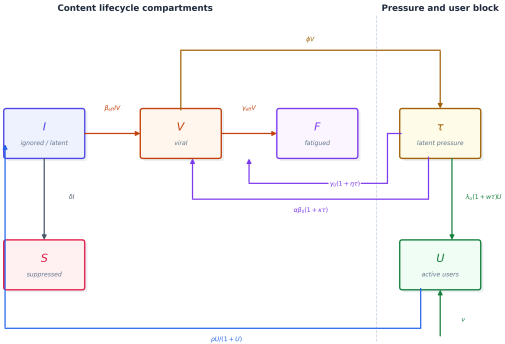
1. A sharp amplification **threshold** exists
2. Above it, a **toxicity feedback loop** locks in
3. The engagement–health **tradeoff is quantifiable**

The IVFS Model

Six state variables:

- I, V, F, S: content lifecycle ignored $\rightarrow$ viral $\rightarrow$ fatigued / suppressed
- τ: latent discussion pressure
- U: active user pool

Key idea: we classify *content*, not users. Platforms rank content; content volumes are directly observable.



Deterministic compartmental ODE. $\beta_{eff} = \alpha\beta_0(1+\kappa\tau)$; latent pressure τ feeds back into spread, fatigue, and user retention.

Threshold and Feedback Loop

Amplification threshold

$$\mathcal{R}_0 = \frac{\alpha\,\beta_0\,I_{\mathrm{DFE}}}{\gamma_0 + \mu_c} \qquad \boxed{\alpha_{\mathrm{crit}} \approx 0.3165}$$

- Below threshold: virality dies out, no pressure
- Above threshold: endemic virality, users leave
- Transition is **transcritical**

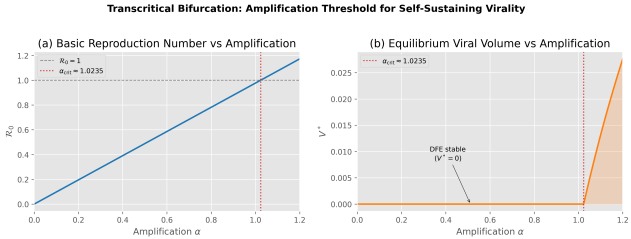
The toxicity feedback loop

$$V \xrightarrow{\phi} \tau \xrightarrow{\kappa} \beta_{\mathrm{eff}} \to V$$

Viral content generates discussion pressure, which amplifies spread, which creates more viral content.

Counteracted by fatigue, pressure decay, and user departure.

Results: Bifurcation and Policy Tradeoff



Key numbers

Moderate ($\alpha = 0.5$) to aggressive ($\alpha = 0.9$):

- Viral volume up 73%
- Discussion pressure up 73%
- User base down 40%

Sensitivity analysis: spread parameters (γ_0, α, β_0) dominate viral volume; pressure parameters (ϕ, w) drive user retention.

Calibration and Limitations

Spread block: well-identified

- Higgs Twitter retweets, 100 h window
- Full-window $R^2 = 0.92$, narrow bootstrap CIs
- Beats SIR benchmark (spike-window $R^2 = 0.89$ vs. 0.69)

Pressure block: partial

- Reply activity as proxy
- Anchored to external toxicity scale
- Qualitative ranking is robust

Limitation: pressure parameters (ϕ, ψ) are not uniquely identified from available data. The spread calibration is strong; the pressure block is proxy-grounded.

Takeaways

1. **Threshold.** A sharp amplification threshold exists ($\alpha_{\text{crit}} \approx 0.3165$). Below it, no sustained virality.
2. **Feedback loop.** Above threshold, a self-reinforcing cycle: viral content $\rightarrow$ pressure $\rightarrow$ amplified spread $\rightarrow$ more viral content.
3. **Tradeoff.** Aggressive amplification buys engagement today at the cost of a smaller, more toxic, less sustainable platform tomorrow.

Thank you. Questions?

Code and report: `github.com/n33levo/apm348_project`

Backup: Full ODE System

$$\dot{I} = \frac{\rho\, U}{1+U} - \beta_{\text{eff}}\, I\, V - (\delta + \mu_c)\, I$$

$$\dot{V} = \beta_{\text{eff}}\, I\, V - \gamma_{\text{eff}}\, V - \mu_c\, V$$

$$\dot{F} = \gamma_{\text{eff}}\, V - \mu_c\, F$$

$$\dot{S} = \delta\, I - \mu_c\, S$$

$$\dot{\tau} = \phi\, V - \psi\, \tau$$

$$\dot{U} = \nu - \lambda_u (1 + w\tau)\, U$$

$$\beta_{\text{eff}} = \alpha\, \beta_0\, (1 + \kappa\, \tau), \qquad \gamma_{\text{eff}} = \gamma_0\, (1 + \eta\, \tau)$$

Backup: Scenario Numbers & Sensitivity

Scenario	α	$\mathcal{R}_0$	V^*	τ^*	U^*
Engagement-first	0.9	2.84	0.1203	0.0674	29.88
Moderate	0.5	1.58	0.0697	0.0390	35.97
Health-first	0.2	0.63	0.0000	0.0000	50.00

	$S(V^*, p)$	$S(\tau^*, p)$	$S(U^*, p)$
α	$+1.67$	$+1.67$	-0.469
β_0	$+1.67$	$+1.67$	-0.469
γ_0	-2.59	-2.59	$+0.726$
ϕ	≈ 0	$+1.00$	-0.281
ψ	≈ 0	-1.00	$+0.281$

Backup: Pressure Configurations

Configuration	ϕ	ψ	Proxy R^2	Moderate τ^*
Current baseline	0.056	0.100	0.18	0.039
Higgs decay-anchored	0.061	0.041	0.10	0.104
External scale	0.146	0.100	0.31	0.102
Reference-constrained	0.417	0.286	0.41	0.102
Higgs (boundary-constrained)†	0.960	1.000	0.74	0.067

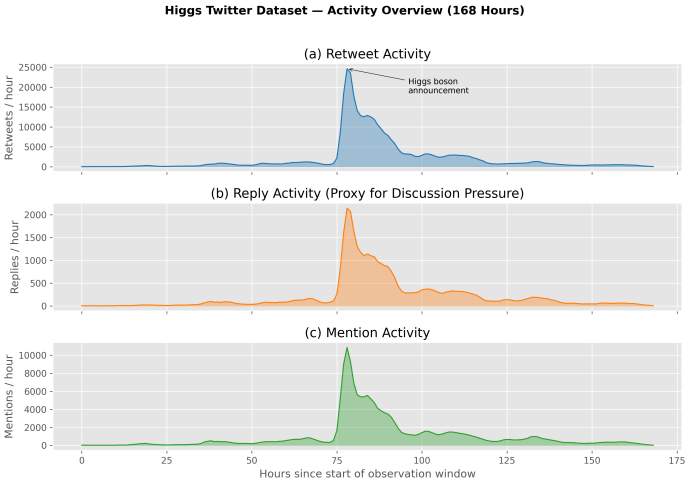
†Optimizer reached upper bound $\psi = 1.0$; true unconstrained optimum may have $\psi > 1$.

The qualitative policy ranking (engagement-first $>$ moderate $>$ health-first for pressure) is preserved across **all five** configurations.

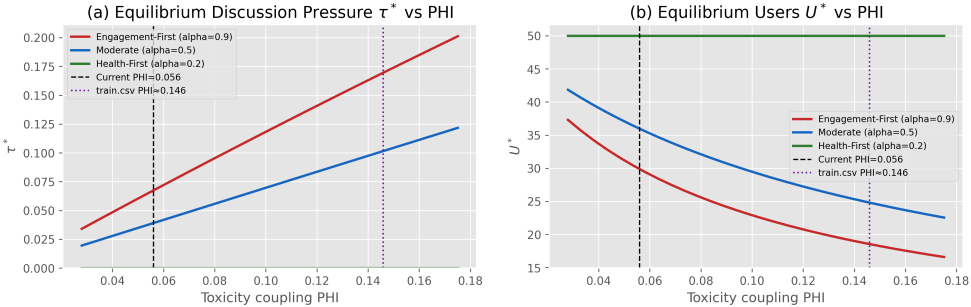
Backup: Well-Posedness

- **Nonnegativity:** Every face of $\mathbb{R}^6_{\geq 0}$ is inward-pointing.
- **Boundedness:** $U \leq \nu/\lambda_u$, total content $C \leq \rho/\mu_c$, $\tau \leq \phi C_{\max}/\psi$.
- **DFE stability:** Jacobian has block-triangular structure; V-eigenvalue is $(\gamma_0 + \mu_c)(\mathcal{R}_0 - 1)$.
- **Positive equilibrium:** Numerically verified LAS at all tested $\alpha > \alpha_{\text{crit}}$.

Backup: Higgs Dataset Overview



Backup: ϕ Sensitivity



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